



INTERNSHIP REPORT

On

“Liver Patient Prediction”

*A report submitted in partial fulfillment of the requirements for the Award of
Degree of*

Bachelor of Technology in Artificial Intelligence & Data Science

at

Dr. Babasaheb Ambedkar Technological University

Lonere, Raighad.



Submitted by

Mane Alankar Vijaykumar

PRN No: 2221301995524

Class: Final B.Tech.

Roll No:42

Under the Guidance of

Prof. J.K. Bhosale



T.P.C.T.'S

College of Engineering, Osmanabad (Dharashiv)

Department of Artificial Intelligence & Data Science

Academic Year 2024-25 (Part-II) Semester VIII



Certificate

This is to certify that the Internship report on “**Liver Patient Prediction**” submitted by **Mane Alankar Vijaykumar (2221301995524 & 42)** is the work done by him/her and submitted during the Academic Year 2024-25 (Part-II) Semester VIII, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Artificial Intelligence and Data Science** at **Dr. Babasaheb Ambedkar Technological University Lonere, Raighad**.

Prof. J.K. Bhosale

Guide

Prof. S.V. Tekale

D.A.C

Prof. P.M. Pawar

HOD (AI&DS)

Dr. V.V. Mane

Principal

DECLARATION

I hereby declare that, I / We Mr./Ms. “Mane Alankar Vijaykumar” is responsible for the work submitted in this project entitled " Liver Patient Prediction " that, the original work is my own except as specified in acknowledgments or in footnotes. To the best of my knowledge and belief, this project report contains no material previously published or written by another person, except where due reference is made and that neither the project report nor the original work contained herein has been submitted to this or any other institution for the award of any degree.

Date:-

Mane Alankar Vijaykumar

Place:-

(2221301995524)

ACKNOWLEDGEMENT

There are to acknowledged and thank all the individuals who played defining role in shaping this project report. Without their constant support, guidance and assistance this report would not have been completed alone. I / We would personally thankful to my / our guide **Prof. J.K. Bhosale** ' for his expert guidance, continuous encouragement and ever willingness to spare time from his busy schedule for the project progress review. I / We am/ are also thankful to our Project coordinator. I / We express sincere thanks to Prof. P.M. Pawar, Head of Artificial Intelligence and Data Science Department for their valuable guidance, support and motivation. I / We am / are also thankful to Dr. V.V. Mane, Principal, TPCT's College of Engineering Dharashiv without whose support and encouragement, the work could not have been completed. I / We would like to thanks all Teaching and Non-teaching staff as well as the helping hands which directly or indirectly helped me during my project work. This work would have not been possible without their valuable help and motivation last but not the least my success and confidence lies solely on the blessing of my / our mother and my / our family.

Mane Alankar Vijaykumar
(2221301995524)



Dr. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE, DIST: RAIGADH.



T.P.C.T.'s
COLLEGE OF ENGINEERING,
DHARASHIV (OSMANABAD).

CERTIFICATION OF COMPLETION

Of Industrial Training

This is to certify that Mr./Ms. **Mane Alankar Vijaykumar** with class: **Final Year B. Tech** with **Enrolment No: 2221301995524**, is currently undergoing an internship at **Rubixe AI Company**.

The internship commenced on **10th March 2025** and is **currently ongoing**, as part of the partial fulfilment towards the completion of the **B.Tech Degree in Artificial Intelligence and Data Science** under the discipline of **T.P.C.T.'s College of Engineering, Dharashiv (Osmanabad)**.

We appreciate his contribution and dedication during the internship period so
Institute Code: 2130

Signature
Head of Department

Seal of Institute

Signature
Principal

STICK THE INDUSTRY CERTIFICATE



+91 (0) 804-717-8999

care@rubixe.com

www.rubixe.com

C-25, Hosur Road, Kudlu Gate,
Bangalore - 560068



Internship Offer Letter

Dear Mane Alankar Vijaykumar ,

Rubixe a brand of Think Ahead Innovations Private Limited is pleased to offer you an educational internship opportunity in Data Science Consultant. This position located in Bajrang House, C-25, Kudlu Gate, Hosur Road,Behind Trident Hyundai Showroom,Bangalore- 560068.

Your Schedule will be approximately 12 hours/ per week, beginning 10-03-2025 and your assignment will conclude on or before 180 days.

Sincerely,

Director



Document No: RBXINTSHPOFF25060517491071068679

DIARY
FOR INTERNSHIP

Name of the Company: Rubixe AI Company

From: 10-March-2025 To 10-September-2025

Name of the Representative: S.Srinivas

Designation of the Representative: Associate Team Lead

Mobile No of the Representative: 9731442527

Name of the Student: Mane Alankar Vijaykumar

Discipline: Artificial Intelligence and Data Science

Signature of the Student:

Signature of Industry representative:

Format 1 Company Profile

Collecting Information about Industry / Organization, available for training along with capacity: -

1. Name of the Industry / Organization: Rubixe AI Company
2. Address / communication details with email: Rubixe, HSR Layout, Bengaluru, Karnataka, India. Email- info@rubixe.com
3. Contact person details:
 - a. Name: S.Srinivas
 - b. Designation: Associate Team Lead
 - c. Email: srinivas.s@datamites.com
 - d. Contact numbers: 9731442527
4. Type of Product:

Service industry / Manufacturing Industry / Other : Artificial Intelligence Solutions and Consulting
5. Type of Control:

Govt. / Semi Govt. / PSU / Pvt. / Prop. : PVT
6. Type of Company:

Large Scale / Medium scale / Small scale : Medium scale
7. Total no. of employees in the Industry / Organization: Approximately 450
8. Details
 - I. Whether willing to offer Industrial training facility during May / June for Degree in Engineering students : Yes / No =Yes
 - II. If yes, whether you offer 6 weeks training : Yes / No III. Internship capacity possible:

Programme engg. group	Civil	Mech	CSE	ETC	AIDS	B Pharmacy
Male			1		2	
Female					1	
Total			1		3	

Name & signature of Industrial Representative.

Format 2

Consent Letter from Parents / Guardians

To,

**The principal,
College of engineering, Osmanabad.**

Subject: Consent for Industrial Training.

Respected Sir,

I am fully aware that –

- I. My ward **Mane Alankar Vijaykumar** studying in VIII semester at your institute has to undergo 24 weeks of internship for partial fulfillment towards completion of Internship.
- II. For this fulfillment he has been deputed at Rubixe AI Industry, located at **Bengaluru, Karnataka** for industrial training of **24 weeks** for the period from **10/03/2025** to **10/09/2025** With respect to above, I give my consent for my ward to travel to & from the mentioned industry.

Further I undertake that-

- a. My ward will be entirely under the discipline of the origination where he / she will be placed & will abide by rules & regulations in force, of the said organization.
- b. My ward is not entitled to any leave during the raining period.
- c. My ward will submit regularly, a prescribed weekly dairy, duly filly & countersigned by the training supervisor of the organization to the mentor faculty of the college.
- d. My ward will undergo the training at his / her own cost & risks during training and / or stay.

I have explained the contents of the letter to my ward who has also promised to adhere strictly to the requirements. I assure that my ward will be properly instructed to take his own care to abide any accidents / injuries in the industry.

Date:

Place:

Signature of Parent/ Guardian

Name:

Address:

Phone No.

Month 1: From 10-March-2025 to 10-April-2025

Expected work:

- Understand project background and scope
- Explore and understand dataset
- Learn related work and research
- Basic cleaning and preparation

Day	Activities carried out
1	Orientation and discussion on project expectations
2	Overview of the dataset: structure, types of features
3	Literature review on similar liver disease prediction problems
4	Setup working environment (Python, Jupyter, libraries)
5	Importing and previewing dataset
6	Initial data cleaning (null values, duplicates)

Weekly summarization of the above activities: The first week involved understanding the scope of the Liver Patient Prediction project, reviewing the dataset, and setting up the development environment. Basic data cleaning was initiated.

Signature of the student

Signature of the industry supervisor

Month 2: From 10-April-2025 to 10-May-2025

Expected work:

- Perform in-depth exploratory data analysis (EDA)
- feature relationships and distributions
- outliers, missing values, encode categorical data
- Start feature selection/engineering

Day	Activities carried out
1	Univariate analysis (histograms, value counts)
2	Bivariate analysis (correlation matrix, scatter plots)
3	Handle missing values, imputation methods
4	Encode categorical features (Label/One-Hot Encoding)
5	Outlier detection and treatment
6	Feature selection and creation of new features

Weekly summarization of the above activities: Second month focused on Performed detailed exploratory data analysis to understand data distributions and relationships. Handled missing values, encoded categorical features, and treated outliers effectively. Completed initial feature selection and engineering for improved model readiness.

Signature of the student

Signature of the industry supervisor

Month 3: From 10-May-2025 to 10-June-2025

Expected work:

- Train baseline ML models
- Evaluate with accuracy, precision, recall, F1-score
- Perform cross-validation
- Select best-performing model(s)

Day	Activities carried out
1	Train-test split of the dataset
2	Train initial models (Logistic Regression, Decision Tree)
3	Train ensemble models (Random Forest, XGBoost)
4	Model evaluation (confusion matrix, accuracy, F1)
5	Apply k-fold cross-validation
6	Select best-performing model for deployment

Weekly summarization of the above activities: Trained multiple machine learning models including Random Forest and XGBoost. Evaluated models using accuracy, precision, recall, and F1-score metrics. Performed cross-validation and finalized the best-performing model for use.

Signature of the student

Signature of the industry supervisor

Month 4: From 10-June-2025 to 10-July-2025

Expected work:

- Advanced data cleaning and preprocessing
- Data transformation and feature scaling
- engineering experimentation

Day	Activities carried out
1	Performed thorough null value handling and imputation techniques
2	Applied feature scaling methods (Standard Scaler, Min Max Scaler)
3	Created new features and transformed existing ones
4	Handled class imbalance (SMOTE/undersampling if applicable)
5	Evaluated feature importance and reduced irrelevant columns
6	Finalized preprocessed dataset for modeling phase

Weekly summarization of the above activities: Focused on model optimization. Handled scaling, transformation, and class imbalance to improve model readiness. Finalized a clean, structured dataset for accurate and reliable model training.

Signature of the student

Signature of the industry supervisor